

Henrique Moura Dias

Research Associate | Plant-Microbe-Environment Interactions & Bioinformatics

Department of Agronomy, Horticulture and Plant Science | South Dakota State University

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RESEARCH INTERESTS

Primary Fields: Plant-Biotic-Abiotic Interactions, Plant-Microbe Coevolution, Multi-Omics Integration.

Technical Expertise: Bioinformatics, Computational Systems Biology, High-Throughput Genomics, Transcriptomics and Metabolomics, and Isotope Enrichment.

Applied Context: Crop Adaptation in Agroecosystems, Plant Stress Physiology, Plant-Growth Promoting Microorganisms, Nutrient Use Efficiency (NUE).

RESEARCH EXPERIENCE

South Dakota State University (SDSU) | Rapid City, SD, USA

Postdoctoral Fellow – Dept. of Agronomy, Horticulture and Plant Science | Aug 2024 – Present

- **Supervisor:** Dr. Christopher Graham
- **Focus:** Leading computational and multi-omics pipelines to evaluate plant-microbe-soil dynamics and genetic adaptations under environmental stress in agroecosystems.

University of São Paulo (USP) | São Paulo, Brazil

Graduate Research Associate – Department of Botany | Feb 2019 – Feb 2024

- **Advisor:** Dr. Marie-Anne Van Sluys
- **Focus:** Investigated functional genomics and genomic stability in plants, focusing on biotic stress response and computational modeling of molecular networks.

University of Nebraska-Lincoln (UNL) | Lincoln, NE, USA

Visiting Graduate Fellow – Center for Plant Science Innovation | May 2023 – Nov 2023

- **Advisor:** Dr. James Schnable
- **Focus:** Implemented advanced quantitative genetics and bioinformatic frameworks for comparative plant genomics during an intensive international doctoral research residency.

Molecularis SA | São Paulo, Brazil

Specialist Scientist (Industry R&D) | May 2022 – May 2023

- **Supervisor:** Dr. Paula Turrini
- **Focus:** Developed high-throughput molecular workflows, managed cross-functional R&D projects, and applied systems biology approaches to commercial biotechnology solutions.

ACADEMIC HISTORY

University of São Paulo | Brazil

2019-2024

Ph.D., Biotechnology (Systems Biology)

Thesis: Functional aspects of TH1 gene in plants (Focus on plant-stress and molecular mechanisms)

University of São Paulo | Brazil

2020-2022

MBA in Project Management

Thesis: Technological trajectory, environmental sustainability, and innovation prospects in the Brazilian sugarcane biofuel sector

Federal University of Technology-Paraná | Brazil

2014-2019

B.S. in Forester Engineering

Thesis: Dynamics and homeostasis of the Tnt1 retroelement and its participation in ethylene pathways in *Nicotiana tabacum*

GRANTS, AWARDS, AND FELLOWSHIPS

Funded

Co-Principal Investigator | SPARC Planning Grant, South Dakota State University | **\$14,969** awarded | 2025

- *Project Title/Focus:* [Insira 1 linha sobre o foco da pesquisa aqui, conectando a dados/ômicas].

Lead applicant | Pulse Crops Working Group – USDA (Federal Support) | **\$1,500** awarded | 2024

- *Deliverable:* Travel and research presentation funding for federal collaborative working group.

Graduate Fellowship (International Residency) | São Paulo Research Foundation (FAPESP), Brazil | **\$19,888** awarded | 2023

- *Purpose:* Funded international research stay at the University of Nebraska-Lincoln.

Graduate Research Fellowship Program | São Paulo Research Foundation (FAPESP), Brazil | **\$43,347** awarded | 2019-2024

- *Purpose:* Funded fellowship research program at the University of São Paulo.

Pending

Principal Investigator | Fertilize During Drought or Wait for Rain? Tracking Nitrogen Recovery in Western South Dakota Spring Wheat. NREC 2027 Research Program. **\$64,800 requested** | 2026 | Under review.

Principal Investigator. Predicting Microbial Resuscitation and Carbon Allocation from Drought Legacy in Rewetted Grassland Soils. DOE Joint Genome Institute, Community Science Program – New Investigator Call FY27. **User-facility resources requested:** 180 environmental metatranscriptomes, 40 polar-metabolomics analyses, and 60 non-polar-metabolomics analyses | 2026 | Status: Under review.

Selected Competitive Proposals Submitted

Principal Investigator. Healthy Diets Equal Healthy Bison: Advancing Bison Dietary Diversity through Forage Improvement Strategies Using Metabarcoding, Chemistry, and Microbiome Profiling. Center of Excellence for Bison Studies. **\$204,616 requested** | 2025 | Status: Not funded.

Principal Investigator. Genomic and Rhizosphere-Microbiome Profiling of Drought-Tolerant Camelina Varieties in the Northern Great Plains. USDA-NIFA. **\$128,690 requested** | 2026 | Status: Not funded.

PEER-REVIEWED PUBLICATIONS

Summary: 12 peer-reviewed publications, including 6 first- or co-first-author articles and 2 additional corresponding-author articles; 1 book chapter; 3 teaching resource; and 5 publicly available bioinformatics pipelines and computational workflows.

De Toledo, NA., Van Sluys, MA., **Dias, HM***. Evolutionary reconstruction of thiamine biosynthesis pathway: An integrative bioinformatics workflow. *Methods in Enzymology*, 2026. DOI: 10.1016/bs.mie.2026.06.014. *Corresponding author.

Dias, HM., Sagawa, GY., Torres-Rodriguez, VJ., Mural, RV., Schnable, JC., and Van Sluys, MA. A sorghum-anchored pan-grass syntenic gene set in grasses. *NAR Genomics and Bioinformatics*, 2026, 8(2). DOI: <https://doi.org/10.1093/nargab/lqag030>.

Dias, HM., Jain, R., Santos, VA., Hernandez-Gonzalez, JLG., Solanki, S., Menendez-III, HM., and Graham, C. High-fidelity soil and plant microbiome profiling with a reproducible Emu workflow on HPC clusters. *Bio-Protocol*, 2026, 16(2). DOI: 10.21769/BioProtoc.5577.

Graham, C., Blair, A., Brennan, J., Cammack, K., **Dias, HM.**, Menendez, H., and Zuidema, D. Soil carbon fraction responses to grazing intensity and texture in a semiarid grassland. *Soil Science Society of America Journal*, 2026, 90, e70184. DOI: <https://doi.org/10.1002/saj2.70184>.

Zerillo, MM., **Dias, HM.**, Ishida, JK., Bonadio, DT., Turrini, PCG., Bini, AP., Miranda, RP., Monteiro-Vitorello, CB., Camargo, LEA., Creste, S., and Van Sluys, MA. In planta transcriptional profile of *Xanthomonas albilineans* brings new insights into its early adaptation to the nutrient-restricted xylem of sugarcane. *Phytopathology*, 2026. DOI: 10.1094/PHYTO-06-25-0201-R.

Dias, HM., Hernandez, JLG., and Graham, C. Influence of plant genotype on nodule-associated bacterial community composition and function in seasonal field pea varieties.” *Legume Science*, 2025, 7(4). DOI: <https://doi.org/10.1002/leg3.70069>.

Partida, VGS.*, **Dias, HM.***, Portes, MT., and Van Sluys, MA. The genetic puzzle of multicopy genes: challenges and troubleshooting. *Plant Methods*, 2025, 21(1), 32. DOI: doi.org/10.1186/s13007-025-01329-0. *Equal contribution.

Seidel, DS., Claudino, PH., Sperotto, G., Wendt, SN., Shomo, ZD., Mural, RV., and **Dias, HM***. Comprehensive analysis of the aquaporin genes in *Eucalyptus grandis* suggests potential targets for drought stress tolerance. *Tropical Plant Biology*, 2024, 17(4), 233–248. DOI: doi.org/10.1007/s12042-024-09364-2. *Corresponding author.

Dias, HM., de Toledo, NA., Mural, RV., Schnable, JC., and Van Sluys, MA. *THI1* Gene evolutionary trends: a comprehensive plant-focused assessment via data mining and large-scale analysis. *Genome Biology and Evolution*, 2024, 16(10), evae212. DOI: <https://doi.org/10.1093/gbe/evae212>.

Sahay, S., Shrestha, N., **Dias, HM.**, Mural, RV., Grzybowski, M., Schnable, JC., and Głowacka, K. Nonphotochemical quenching kinetics GWAS in sorghum identifies genes that may play conserved roles in maize and *Arabidopsis thaliana* photoprotection. *The Plant Journal*, 2024, 119(6), 3000–3014. DOI: <https://doi.org/10.1111/tpj.16967>.

Dias, HM., Vieira, AP., de Jesus, EM., de Setta, N., Barros-Carvalho, GA., Van Sluys MA. Functional and comparative analysis of *TH1* gene in grasses with focus on sugarcane. *PeerJ*, 2023, 11, e14973. DOI: <https://doi.org/10.7717/peerj.14973>.

Partida, VGS., **Dias, HM.**, Corcino, DSM., and Van Sluys., MA. Sucrose-phosphate phosphatase from sugarcane reveals an ancestral tandem duplication. *BMC Plant Biology*, 2021, 21(1), 23. DOI: <https://doi.org/10.1186/s12870-020-02795-5>.

Cerqueira, FG., Souza, ACSS. de, Thompson, BB., Michalichen, G., Brito, GGS., Pereira, MOF., Kreczkuski, C., Canonico, CM., Porrua, DA., Biolchi, G., **Dias, HM.**, Arancibia, IAL., Ribeiro, JCB., Minozzo, M., Reis, MEM., Felipe, NA., Krefta, SC., Potrich, M., and Tessaro, D. Environmental education - perceptions and reflections based on the project "Plante Essa Ideia". *Research, Society and Development*, 2020, 9(9), e746997710. DOI: <https://doi.org/10.33448/rsd-v9i9.7710>

MANUSCRIPTS IN REVIEW/PREPARATION

Jain, R., **Dias, HM.**, Solanki, S., Gonzalez-Hernandez, J., Ameen, G., Graham, C. Stability of the Wheat Rhizosphere Microbiome Across Nitrogen Fertilization Rates Is Consistent with Limited Yield Response in a Semi-Arid System (under review in *Agrosystems, Geosciences and Environment*).

Dias, HM., Viridis, ML., Petronella, A., Muller, LM., Feist, HK., Graham, C., and Menendez, HM. Soil origin and microbiota disturbance differentially shape bacterial and fungal root-associated communities during native grass establishment (under review in *Soil Ecology Letters*).

Dias, HM., Muller, LM., Hernandez-Gonzalez, JLG., Menendez-III, HM and Graham, C. A short-term glucose pulse reveals decoupling between soil microbial community structure and transcriptional dynamics in grassland soil (under review in *Biology and Fertility of Soils*).

Dias, HM., Anataya, A., Husmann, A., Feist, H., Parsons, I., Zuidema, D., Cammack, K., Blair, A., Brennan, J., Martin, J., Short, R., Elert, K., Gonzalez-Hernandez, J., Menendez, H., Graham, C. Multi-scale microbiome datasets from an 80-year grazing experiment in Northern Great Plains grasslands (under review in *Scientific Data*).

Leite-Junior, JN., Cirino, H., Antonello, P., Zerillo, MM., **Dias, HM.**, and Van Sluys, MA. Unmapped RNA-Seq reads from inoculated sugarcane reveals long non-coding RNAs related to retrotransposons and suggest microbiome modulation (under review in *Genetics and Molecular Biology*). Biorxiv: doi.org/10.1101/2025.09.28.679069.

Hernandes-Lopes, J., Quintanilha, DM., Jesus, EM., Lopes, FM., Parmigiani, RB., Karolski, B., **Dias, HM.**, Jacobs, TB., Camargo, AA., Kitajima, JP., Van Sluys, MA. Evidence-based gene expression modulation correlates with transposable element knock-down (under review in *Scientific Reports*). Biorxiv: doi.org/10.1101/2020.08.15.252312.

Dias, HM.*, Gonzalez-Hernandez, J., Graham, C. Metatranscriptome mining reveals diverse RNA viruses and mitochondrial-code signatures of fungal mitoviruses in grassland soil (in advanced preparation).
*Corresponding author

Dias, HM., Leite-Junior, JN., Bonadio, DT., Sagawa, GYA., de Toledo, NA., Corcino, DSM., Partida, VGS., Berselli, AP., Mural, RV., Grandis, A., Buckeridge, MS., Pisetta, TCS., Creste, S., and Van Sluys, MA. Drought-induced carbon remobilization is associated with delayed recovery of thiamine biosynthetic gene expression in sugarcane. 2025 (in advanced preparation).

Dias, HM.* and Graham, C. Molecular signatures of coupled biogeochemical cycling in soil microbiome (in advanced preparation). *Corresponding author

INVITED TALKS AND SEMINARS

- **Dias, H. M.** (2026). *Resident Soil Microbial Structure Stays Put While the Active Community Responds to a Carbon Pulse*. Invited Speaker, CANVAS 2026, Portland, Oregon, USA.
- **Dias, H. M.** (2025). *An introduction to bioinformatics in 16S rRNA microbiome analysis*. Guest Instructor & Invited Seminar, Training Workshop, Oglala Lakota College, South Dakota, USA.
- **Dias, H. M.** (2025). *Microbes can drive your life, and I can prove it!* Invited Departmental Seminar, West River Research Center, South Dakota State University, USA.
- **Dias, H. M.** (2024). *Microbial dynamics in rhizobia nodules of seasonal yellow pea varieties*. Invited Research Presentation, Pulse Crops Working Group (Virtual Meeting), USA.
- **Dias, H. M.** (2022). *Pathogen infection and drought stress regulate genes of the thiamine de novo biosynthesis and salvage pathway in sugarcane*. Invited Talk, Brazilian Society of Genetics, São Paulo, Brazil.
- **Dias, H. M.** (2022). *The battle for survival: plants vs. pathogens*. Invited Institutional Lecture, Institute of Bioscience, University of São Paulo, São Paulo, Brazil.
- **Dias, H. M.** (2022). *Transcriptomics: from experimental planning to analysis*. Invited Technical Workshop, Institute of Biomedical Science, University of São Paulo, São Paulo, Brazil.
- **Dias, H. M.** (2022). *Uncovering hidden biodiversity through sequence analysis and database searches*. Invited Guest Lecture, Department of Molecular Sciences, University of São Paulo, São Paulo, Brazil.
- **Dias, H. M.** (2020). *Long intergenic non-coding RNAs (lincRNAs) associated to adaptive stress in plants*. Invited Departmental Seminar, Institute of Bioscience, University of São Paulo, São Paulo, Brazil.
- **Dias, H. M.** (2019). *Physcomitrella patens: a new model chassis for functional gene characterization*. Invited Lecture, Institute of Bioscience, University of São Paulo, São Paulo, Brazil.

CONTRIBUTED CONFERENCE TALKS AND SELECTED POSTERS

Selected Oral Presentations (Talks)

- **Dias, H. M.**, Viridis, ML., Petronella, A., Muller, LM., Feist, HK., Graham, C., and Menendez, HM. (2026). *Soil legacy and plant-driven resilience of bacterial guilds in grazed grasslands*. Microbes in Microbiomes, Microbial Society. Miami, Florida, USA.
- **Dias, HM.**, and Graham, C. (2025). *Microbial dynamics in rhizobia nodules of seasonal yellow pea varieties*. Pulse Crops Working Group Annual Meeting. [Virtual Meeting].
- **Dias, HM.**, Zapelão, K., Quintino, LF., Leite-Junior, JN., and Van Sluys, MA. (2022). *Pathogen infection and drought stress regulate genes of the thiamine de novo biosynthesis and salvage pathway in sugarcane*. International Congress of the Brazilian Genetics Society. Águas de Lindoia, São Paulo, Brazil.
 - **Award: Honor Merit for Best Oral Presentation (Plant Genetics)**

- De Toledo, NA., **Dias, HM.**, and Van Sluys, MA. (2022). *Analysis of TH11 gene reveals motifs that bring clues of its evolutionary history in Archaea and Eukarya*. International Congress of the Brazilian Genetics Society. Águas de Lindoia, São Paulo, Brazil.
 - **Award: Honor Merit for Best Oral Presentation (Evolution)**

Selected Poster Presentations

- Zerillo, MM., **Dias, HM.**, Ishida, JK., Bonadio, DT., Turrini, PCG., Bini, AP., Miranda, RP., Monteiro-Vitorello, CB., Camargo, LEA., Creste, S., and Van Sluys, MA. (2025). *In planta transcriptomics of Xanthomonas albilineans reveals early adaptations to the nutrient-limited xylem environment of sugarcane*. Brazilian Conference of Microbiology. Aracaju, Sergipe, Brazil.
- Sperotto, G., Claudino, PH., Lovison, I., Wendt, SN., and **Dias, HM.** (2022). *Genomic and transcriptional study of the CDPK (Ca²⁺-dependent protein kinase) gene family in Eucalyptus grandis*. Brazilian Congress of Genetic Resources. [Virtual Meeting].
- **Dias, HM.**, and Van Sluys, MA. (2021). *Dissecting cis-regulatory elements in promoter region of the TH11 gene of sugarcane and comparative analysis of homologs in C4 Poaceae*. Plant Biology, ASPB21. [Virtual Meeting].
- **Dias, HM.**, and Van Sluys, MA. (2019). *Analysis of cis-acting elements of TH11 gene in Arabidopsis thaliana and Physcomitrella patens provide clues about their functions*. International Congress of the Brazilian Genetics Society. Águas de Lindoia, São Paulo, Brazil.
- **Dias, HM.**, Lopes-Filho, JH., Perseguini, JM KC., and Van Sluys, MA. (2018). *Dynamics of the Tnt1 retroelement and circadian clock genes in homeostasis conditions in Nicotiana tabacum*. International Congress of the Brazilian Genetics Society. Foz do Iguaçu, Paraná, Brazil.
- **Dias, HM.**, Sartor, KLC., and Perseguini, JM KC. (2018). *Characterization of microsatellite (SSR) markers identified in QTL regions associated with anthracnose resistance in common bean*. Scientific and Technological Initiation Seminar of UTFPR. Apucarana, Paraná, Brazil.

MENTORING EXPERIENCE

Dayana S. Seidel (Undergraduate Student, Federal University of Technology-Paraná) | 2023 – 2024

- *Project:* Comprehensive analysis of the aquaporin (AQP) genes in *Eucalyptus* sp.
- *Outcome:* **Perr-reviewed publication** in *Tropical Plant Biology*.

Naiara A. de Toledo (Undergraduate Student, University of São Paulo) | 2020 – 2021

- *Project:* Regulation of the vitamin pathway through riboswitches: the case of post-transcriptional control by the TPP-riboswitch in the thiamine pathway.
- *Outcome:* **Awarded Honor Merit** for work presentation at the 67th Brazilian Genetic Society Conference.

Kevyn Zapelão (Undergraduate Student, Federal University of Technology-Paraná) | 2020 – 2021

- *Project:* Potential lincRNAs associated with mechanisms of response to infection by *Xanthomonas oryzae* pv. *oryzae* in *Oryza sativa*.
- *Outcome:* **Awarded Innovative Work Undergraduate Research Fellowship**.

Larissa F. Quintino (Undergraduate Student, Federal University of Technology-Paraná) | 2020 – 2021

- *Project:* Understanding biological mechanisms of lncRNA in *Apis mellifera*.
- *Outcome:* **Awarded Innovative Work Undergraduate Research Fellowship**.

Gabriela Sperotto (Undergraduate Student, Federal University of Technology-Paraná) | 2020 – 2021

- *Project:* Exploratory analysis of genes responsible for lignin production in *Eucalyptus grandis* under biotic and abiotic stress conditions using bioinformatics tools.
- *Outcome:* **Awarded Innovative Work Undergraduate Research Fellowship.**

Ivania Lovison (Undergraduate Student, Federal University of Technology-Paraná) | 2020 – 2021

- *Project:* Characterization of ABA biosynthesis and catabolism genes related to abiotic stress in *Eucalyptus grandis*.
- *Outcome:* **Awarded Innovative Work Undergraduate Research Fellowship.**

Paulo H. Claudino (Undergraduate Student, Federal University of Technology-Paraná) | 2020 – 2021

- *Project:* Comprehensive analysis of the *CDPK* gene family (Ca^{2+} -dependent protein kinase) in *Eucalyptus grandis*.
- *Outcome:* **Manuscript in advanced preparation.**

TEACHING EXPERIENCE AND TRAINING

Instructional & Workshop Leadership

- **Guest Instructor** | Oglala Lakota College, South Dakota, USA | 2025
 - *Course:* Full-length 16S rRNA microbiome bioinformatics workflow. Developed curriculum and led training for hands-on computational biology workflows.
- **Graduate Teaching Assistant** | Department of Molecular Sciences, University of São Paulo, Brazil | 2023
 - *Course:* Molecular Biology Laboratory (CM2015). Provided laboratory guidance, experimental setup support, and conceptual feedback to undergraduate students.
- **Graduate Teaching Assistant** | Department of Botany, University of São Paulo, Brazil | 2018
 - *Course:* Use of model plants to functional gene studies. Assisted in course design and practical lab sessions using model plant systems.

Guest Lectures & Institutional Course Support

- **Co-Instructor / Guest Lecturer** | Federal University of Technology-Paraná (UTFPR), Brazil
 - *Topics of Molecular Biology (EEBB)* | 2020
 - *Plant biotechnology methods: understanding plant-microbe interactions* | 2020
 - *Introductory topics of Bioinformatics* | 2021
 - *A Bioinformatics Pipeline to Explore Transcriptional Regulation* | 2019

PROFESSIONAL SERVICE & SYNERGISTIC ACTIVITIES

Ad-Hoc Peer Reviewer

- *Bioinformatics* | 2026 – Present
- *Virulence* | 2026 – Present
- *BMC Plant Biology* | 2025 – Present
- *Bio-protocol* | 2025 – Present

Professional Affiliations & Memberships

- **American Society for Microbiology (ASM)** | Member, 2024 – Present

- **Microbiology Society** | Member, 2024 – Present
- **Crop Science Society of America** | Member, 2026 – Present

PROFESSIONAL REFERENCES

Dr. Christopher Graham

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 South Dakota State University
 711 N Creek
 Rapid City, SD 57703, USA
 e-mail: christopher.graham@sdstate.edu

Dr. Marie-Anne Van Sluys

PhD. Advisor
 University of Sao Paulo
 270 Rua do Matão, Butantã
 Sao Paulo, SP 05508-090, Brazil
 e-mail: mavsluys@usp.br

Dr. Hector M. Menendez

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Dr. Ashlynn Smith

Collaborator
 Louisiana Tech University
 Ruston, Louisiana
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Dr. Jose Gonzalez-Hernandez

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 South Dakota State University
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